

CALL FOR PAPERS | *Physiology and Pharmacology of Temperature Regulation*

Defending body mass during food restriction in *Acomys russatus*: a desert rodent that does not store food

Roe Gutman, Itzhak Choshniak, and Noga Kronfeld-Schor

Department of Zoology, Tel Aviv University, Tel Aviv, Israel

Submitted 3 March 2005; accepted in final form 3 November 2005

Gutman, Roe, Itzhak Choshniak, and Noga Kronfeld-Schor. Defending body mass during food restriction in *Acomys russatus*: a desert rodent that does not store food. *Am J Physiol Regul Integr Comp Physiol* 290: R881–R891, 2006. First published November 10, 2005; doi:10.1152/ajpregu.00156.2005.—Golden spiny mice, which inhabit rocky deserts and do not store food, must therefore employ physiological means to cope with periods of food shortage. Here we studied the physiological means used by golden spiny mice for conserving energy during food restriction and refeeding and the mechanism by which food consumption may influence thermoregulatory mechanisms and metabolic rate. As comparison, we studied the response to food restriction of another rocky desert rodent, Wagner's gerbil, which accumulates large seed caches. Ten out of 12 food-restricted spiny mice (resistant) were able to defend their body mass after an initial decrease, as opposed to Wagner's gerbils ($n = 6$). Two of the spiny mice (nonresistant) kept losing weight, and their food restriction was halted. In four resistant and two nonresistant spiny mice, we measured heart rate, body temperature, and oxygen consumption during food restriction. The resistant spiny mice significantly ($P < 0.05$) reduced energy expenditure and entered daily torpor. The nonresistant spiny mice did not reduce their energy expenditure. The gerbils' response to food restriction was similar to that of the nonresistant spiny mice. Resistant spiny mice leptin levels dropped significantly ($n = 6$, $P < 0.05$) after 24 h of food restriction, and continued to decrease throughout food restriction, as did body fat. During refeeding, although the golden spiny mice gained fat, leptin levels were not correlated with body mass ($r^2 = 0.014$). It is possible that this low correlation allows them to continue eating and accumulate fat when food is plentiful.

body mass; food restriction; leptin; torpor; *Acomys russatus*

FOOD AVAILABILITY AND QUALITY in desert habitats is spatially and temporally unpredictable, and animals often face periods of food shortage (6, 42, 49). The length of time for which an animal is able to withstand food shortage (endurance time) is a function of the available energy and the rate at which it is used (37). To increase the available energy, some (mainly small) desert mammals use seed caches (reviewed in Ref. 17; also see Refs. 26, 28 and 29), whereas other mammals (mainly large) store energy as body fat (21), which they consume during periods of food shortage. The problem of food shortage is more pronounced in small mammals such as rodents because of their relatively high specific metabolic rate (energy demand/body mass unit; 18), and even more so in desert rodents that do not

create food stores. These rodents have to cope with food shortage periods by physiological means.

One such example is the golden spiny mouse (*Acomys russatus*, *Muridae*) that inhabits rocky deserts in Jordan, Sinai (Egypt), and the south of Israel (24). This rodent does not dig burrows but inhabits rock crevasses and does not store food (41), possibly because its diet is comprised mainly of arthropods (19), which cannot be stored. A recent study showed that arthropod abundance in the golden spiny mouse natural habitat fluctuates on both seasonal and daily scales (46, 47). Therefore, we hypothesized that golden spiny mice are exposed to food shortage periods and must withstand these periods by physiological means, i.e., reduce their rate of energy usage and/or increase their body energy storage, thereby prolonging their endurance time. In mammals, most of the body's fuel reserves are stored in white adipocytes as triglycerides that are burned off during fasting and starvation. The rate of usage of these reserves during food shortage is determined by the energy required for survival. The major energetic requirements, for which energy may become depleted during food shortage periods, are energy needed to sustain life in a resting individual [resting metabolic rate (RMR)], energy used for thermoregulation, and energy spent on activity.

Energy demands of desert mammals are much lower than those of nondesert mammals. Both RMRs and field metabolic rates of desert mammals are 10–30% lower (e.g., 6 and 42) than the value expected from their body mass (18). A further dramatic reduction in energy expenditure as a result of food restriction is well documented in several desert mammal species, including golden spiny mice (e.g., 5, 25, and 42). However, how this reduction is achieved is still unclear (but see Ref. 25). In the present work, we studied the physiological means employed by golden spiny mice for conserving energy during food shortage and refeeding periods and examined the mechanism by which food consumption may influence thermoregulatory mechanisms and metabolic rate. To test our hypothesis that diet plays a role in the evolution of adaptation to food shortage, we studied the response of another rocky desert rodent, Wagner's gerbil (*Gerbillus dasyurus*, *Gerbillidae*), to food restriction. Wagner's gerbil is found in deserts and temperate zones (24); it digs its burrows under rocks, feeds mainly on seeds and leaves, but also invertebrates, and accumulates large seed caches, which it consumes during food shortage periods (16, 39). We predicted that the responses of

Address for reprint requests and other correspondence: N. Kronfeld-Schor, Dept. of Zoology, Tel Aviv Univ., Tel Aviv 69978, Israel (e-mail: nogaks@tauex.tau.ac.il).

The costs of publication of this article were defrayed in part by the payment of page charges. The article must therefore be hereby marked "advertisement" in accordance with 18 U.S.C. Section 1734 solely to indicate this fact.

these two species to food restriction would differ, according to their particular life histories; although both species occupy the same habitat, the golden spiny mouse, unlike Wagner's gerbil, does not store food and therefore should be more adapted physiologically to the fluctuations in food availability.

METHODS

Animals. A breeding colony of golden spiny mice originally trapped near the Dead Sea, Israel, is kept at the Meier I. Segals Garden for Zoological Research at Tel Aviv University (permit number 2003/16295). Wagner's gerbils, trapped in the Ramon crater, Israel, and raised in a breeding colony, were kindly given to us by I. R. Khokhlova and B. R. Krasnov of the Ramon Science Center (Mizpe Ramon, Israel).

During the experiments, the animals were housed in standard plastic cages ($21 \times 31 \times 13$ cm for golden spiny mouse and $17.5 \times 28 \times 13$ cm for Wagner's gerbil) in a temperature-regulated room [$30 \pm 1^\circ\text{C}$ for both species, which is around the low critical temperature of the thermoneutral zone of both species; golden spiny mouse (43) and Wagner's gerbil (15)] under a 12:12-h light-dark cycle. Water and rodent chow (Koffolk serial no. 19510) were given ad libitum or as specified in the experimental protocol.

Body mass and food intake. Body mass of 12 golden spiny mice [66.2 g (SD 12.1)] and 6 Wagner's gerbils [24.85 g (SD 0.85)], all mature males aged between 1 and 2 yr, was measured at least two times a week using electronic scales [Sartorius (± 0.1 g) or Sekel (± 0.01 g)]. To measure individual ad libitum food and energy intake, the mice were given weighed [Sekel (± 0.01 g)] commercial rodents pellets. Leftovers and feces were collected after 48 or 72 h, dried to a constant mass at 60°C , and weighed. Energy content of the pellets (19.3 kJ/g) and feces (for calculating percentage of digestibility) were measured using a bomb calorimeter (Gallenkamp) calibrated by ascending mass of benzoic acid (Analar, 26.45 kJ/g).

During food restriction, an individual daily portion (50% of the individual ad libitum consumption) was given 15 min after onset of dark, when core body temperature peaked under ad libitum food availability in both species (see Fig. 5). Based on a previous work on food restriction in golden spiny mice (31), where two out of six individuals tested died from starvation after losing 30% of their body mass, we decided to define a loss of $>25\%$ of the initial body mass as one possible endpoint for the food restriction period. Another possible endpoint was that of achieving a new constant body mass, which was defined as an insignificant change ($P > 0.05$) of the effect of food restriction on body mass for at least three consecutive measurements separated by 2–3 days.

Body composition. Fat mass was measured one time per week, using Double X-ray Absorption (DXA; Lunar Piximus II). Because it was impossible to measure the transmitter-implanted mice, only the six golden spiny mice that were not implanted were scanned. During the measurements, mice were anesthetized with isoflurane (Rhodic) using an anesthetic machine (Ohmeda) with medical-grade oxygen (2–3% vol). Because this is the first time (as far as we know) that the DXA method has been performed on golden spiny mice, we validated the use of DXA for evaluating body fat content following Nagy and Anne-Laure (26). In short, the fat content of five golden spiny mice was measured using DXA, and the animals were killed by decapitation under anesthesia. The carcasses were divided into $\sim 1\text{ cm}^2$ pieces and dried to a constant weight at 60°C . Fat mass was separated from dried lean body mass by chemical extraction for 24 h in a Soxhlet apparatus using 3:1 Petroleum ethanol-ether (Sigma). We found a linear correlation ($r^2 = 0.92$) and a lack of significant difference between fat content by chemical extraction and by DXA estimation (Paired t -test = 0.18).

Telemetry. Core body temperature and heart rate of six golden spiny mice and body temperature of six gerbils were monitored continually using a Data Science International telemetry system

[transmitter model no. TA10ETA-F20, 3.5 g, (golden spiny mice) or Sirtrack single-stage transmitters, 3.8 g, and RX-900 Televilt scanner-receiver (gerbils)]. Before implantation, DSI transmitters were tested for precision ($\pm 0.01^\circ\text{C}$). Calibration curves ($\pm 0.1^\circ\text{C}$) were constructed for each Sirtrack single-stage transmitter. Mice were anesthetized with isoflurane in medical-grade oxygen using an anesthetic machine (Ohmeda), and the transmitters were implanted in the abdominal cavity. The abdominal wall was sutured with nonabsorbable surgical sutures, and the skin was sutured with absorbable surgical sutures, using a cutting needle (5–0; Dexon). The incision was treated with topical antibiotic (1% silver sulfadiazine; Silverol Cream). Antibiotics (5% Baytril, 24 mg/kg Bayer) were injected intramuscularly before implantation as a prophylactic treatment. At least 1 wk was allowed for recovery before initiation of measurements. Data were collected at 3-min intervals throughout the experiment. Daily average, maximal (highest 9-min interval), and minimal (lowest 9-min interval) heart rate and core body temperature were calculated.

Energy expenditure. Oxygen consumption rates ($\dot{V}\text{O}_2$, $\text{ml O}_2 \cdot \text{g}^{-1} \cdot \text{h}^{-1}_{\text{STPD}}$), as an indirect indicator for energy expenditure, were measured in six Wagner's gerbils and in the six transmitter-implanted golden spiny mice in an open system following Depocas and Hart (8). $\dot{V}\text{O}_2$ was calculated using Withers Eq. 4 (50): $\dot{V}\text{O}_2 = [\dot{V}_E \times (\text{FI}_{\text{O}_2} - \text{F}'_{\text{EO}_2})] / (1 - \text{FI}_{\text{O}_2})$, where $\dot{V}_E$ is the rate of airflow out of the cage ($\text{ml/min}_{\text{STPD}}$), FI_{O_2} is the fractional concentration of O_2 entering the cage, and F'_{EO_2} is the fractional concentration of O_2 of outflowing air. STPD is standard temperature (0°C), pressure (760 mmHg), and dry air. Oxygen consumption rate was measured in the individual cage by covering the cage with a transparent Perspex plate with two holes. Airflow through the cages was 400 ml/min (for golden spiny mice under ad libitum food regime and 350 ml/min for food restriction) or 230 ml/min (gerbils) and was continuously monitored using a Mc-Millan flow meter (± 1 ml/min).

Sampled air was desiccated (Silica gel blue; Fluka), CO_2 was absorbed (Soda lime; Merck), and F'_{O_2} was measured using a paramagnetic O_2 analyzer (Taylor Servomax $\pm 0.001\%$). Flow meters were calibrated before every measurement using dry air (without CO_2) at different flow rates through a calibration cylinder (model 1053; Brooks). The O_2 analyzer was calibrated before every measurement using nitrogen vs. dry air. $\dot{V}_E$ and FO_2 were sampled continuously, averaged, and saved every 30 s. Oxygen consumption was measured in both species every week for a period of 48 h, from which the average daily metabolic rate (ADMR) and RMR (the lowest 20 min of the day) were calculated.

Plasma leptin concentrations. Blood samples were collected from the infraorbital sinus of six Wagner's gerbils and the six X-ray-scanned golden spiny mice every week between 1000 and 1200 using an EDTA-covered microhematocrit capillary tube (Mudulohm). Antibiotic eye drops (Dexamycin; Teva) were applied to the eyes after each sampling. Samples were centrifuged, and plasma was stored at -70°C until analysis. Plasma leptin concentration was measured using a commercial multispecies leptin RIA kit (Linco). The RIA was validated for the two species: leptin-like immunoreactivity in spiny mice and Wagner's gerbils plasma diluted in parallel with purified recombinant human leptin in the multispecies leptin RIA ($r^2 = 0.9907$, $P < 0.001$ for spiny mice and 0.9985 , $P < 0.001$ for Wagner's gerbils), and recovery of exogenous human leptin in spiny mice and Wagner's gerbils plasma was close to 100%. Our within-assay coefficients of variation were 5.18–6.78%, and the between-assay coefficients of variation were 3.08–14.97%.

At the end of the experiment, we surgically removed the transmitters from all animals and returned them to the breeding colony at the Meier I. Segals Garden for Zoological Research at Tel Aviv University. All procedures were conducted in accordance with the Institutional Animal Ethics Committee (L-02–45).

Statistical analysis. Statistical analysis was performed using STATISTICA 6 (StatsSoft). Repeated-measures ANOVA was used to detect and compare the effect of food restriction and refeeding on the

different parameters measured in the tested groups (within factor days at food restriction or refeeding; between factor, where appropriate: species or type of golden spiny mice). Significant ANOVAs were followed by Fisher's least-significant difference post hoc test. Significance level was defined as $P < 0.05$.

RESULTS

Food consumption and energy intake. Ad libitum food consumption was 2.24 (SD 0.33) g/day for golden spiny mice and 2.09 (SD 0.22) g/day for Wagner's gerbils. Based on Grodzinski and Wunder (14), we assumed that the metabolized energy intake (MEI) at ad libitum was 2% lower than digestible energy intake. Digestibility in golden spiny mice did not change significantly between ad libitum and food restriction ($P > 0.05$); therefore, we can assume that MEI was reduced by 50%, in accordance with the reduction in food intake. Following Degen and Kam (7), we assumed that the digestible energy intake of Wagner's gerbils was 90.5% of their gross energy intake. Hence, MEI of golden spiny mice and Wagner's gerbils during ad libitum were 38.5 (SD 1.65) and 36.1 (SD 3.72) kJ/day, respectively.

Body mass and composition. Ten golden spiny mice (henceforth, resistant) lost $\sim 0.7\%$ of their body mass per day during the first 12 days of food restriction, reaching $\sim 91\%$ of their

average ad libitum body mass during that period. Their body mass stabilized after 35 days of food restriction ($P > 0.05$ during three consecutive measurements during food restriction, Fig. 1A), after losing $\sim 15\%$ [10.7 (SD 4.9) g] of their average ad libitum body mass. Once body mass had stabilized, the animals received food ad libitum. In two golden spiny mice individuals (henceforth, nonresistant), body mass dropped $>25\%$ [14.3 (SD 1.8) g] in 12 days ($\sim 2.26\%/day$), food restriction was stopped, and their data were analyzed separately. The rate and magnitude of weight loss was higher in Wagner's gerbils ($\sim 1.36\%/day$) than in the resistant golden spiny mice (days at food restriction, $P < 0.001$; species, $P < 0.01$; interaction $P < 0.001$, Fig. 1A). Body mass of the gerbils dropped continuously, and food restriction was therefore stopped after 18 days, when they had lost $\sim 25\%$ [6.2 (SD 1.2) g] of their average ad libitum body mass (Fig. 1A). The reduction of body mass of golden spiny mice resulted mainly from their use of fat stores, as indicated by the decrease in body fat during food restriction [8.7 g (SD 4.9), Fig. 1C].

At the end of the refeeding period, both species regained weight ($P < 0.05$ for both species). After 26 days, body mass of golden spiny mice had reached their ad libitum body mass, whereas the gerbils had not reached their ad libitum body mass

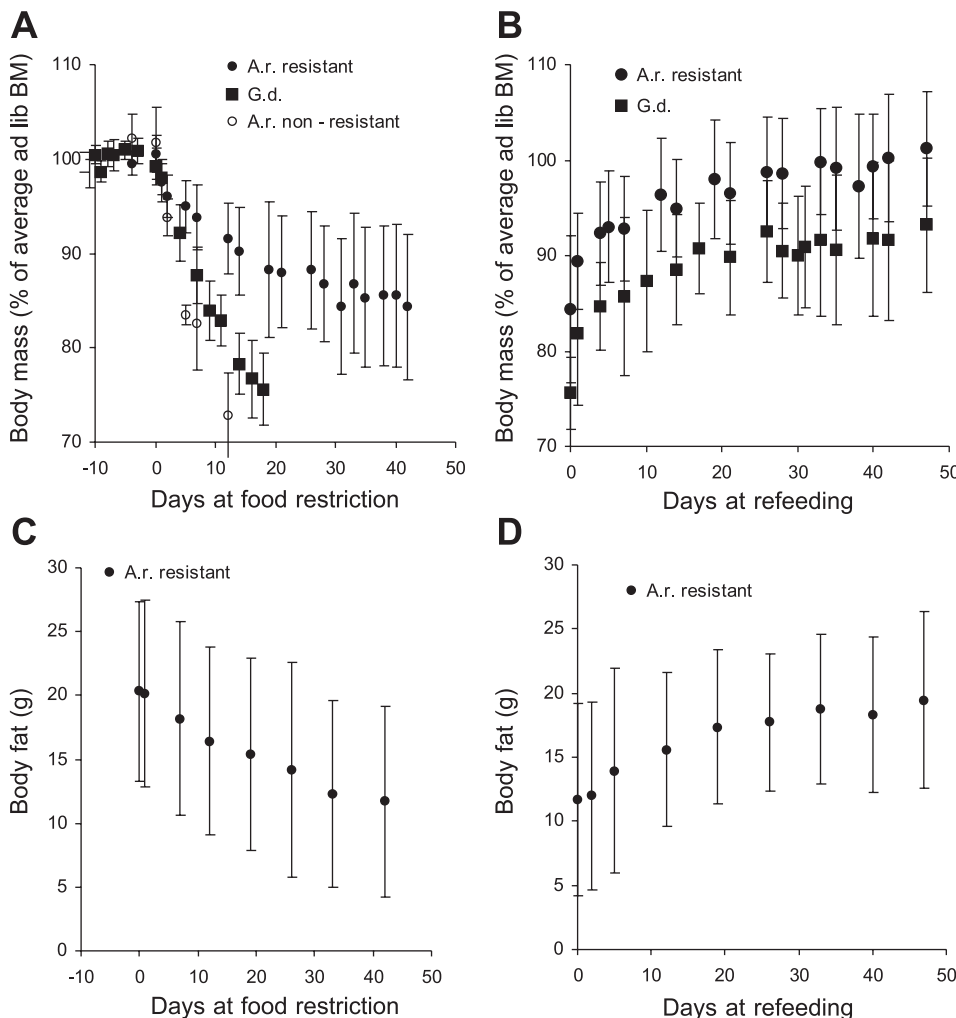


Fig. 1. Changes in body mass (mean $\pm$ SD) during food restriction (A) and refeeding (B) in resistant ($n = 10$) and nonresistant ($n = 2$) golden spiny mice (A.r.), and Wagner's gerbils (G.d., $n = 6$). Values are relative to average body mass at ad libitum [A.r. resistant, 67.6 (SD 3.84), A.r. nonresistant 57.4 (SD 10.35), G.d. 24.9 (SD 0.85)], and changes in body fat ($n = 6$, mean $\pm$ SD) during food restriction (C) and refeeding (D) in resistant golden spiny mice.

even after 47 days of refeeding (post hoc analysis, $P < 0.05$ from day 26 and on for golden spiny mice, $P > 0.05$ for all measurements during refeeding, Fig. 1B). The increase in body mass of golden spiny mice during refeeding resulted mainly from an increase in fat stores, as indicated by the increase in body fat [7.7 g (SD 6.7), Fig. 1D].

Metabolic rate. ADMR of resistant golden spiny mice decreased significantly after 2 days of food restriction ($P <$

0.001; post hoc analysis between ad libitum and 2 days of food restriction, $P < 0.001$, Fig. 2A). It continued to decrease significantly for another week (post hoc analysis between ad libitum and 1st wk of food restriction and between 2 days and 1 wk at food restriction, $P < 0.001$), after which it remained significantly low for the rest of the food restriction (post hoc analysis between ad libitum and all days of food restriction, $P < 0.001$ and $P > 0.05$ between 1 wk at food restriction and

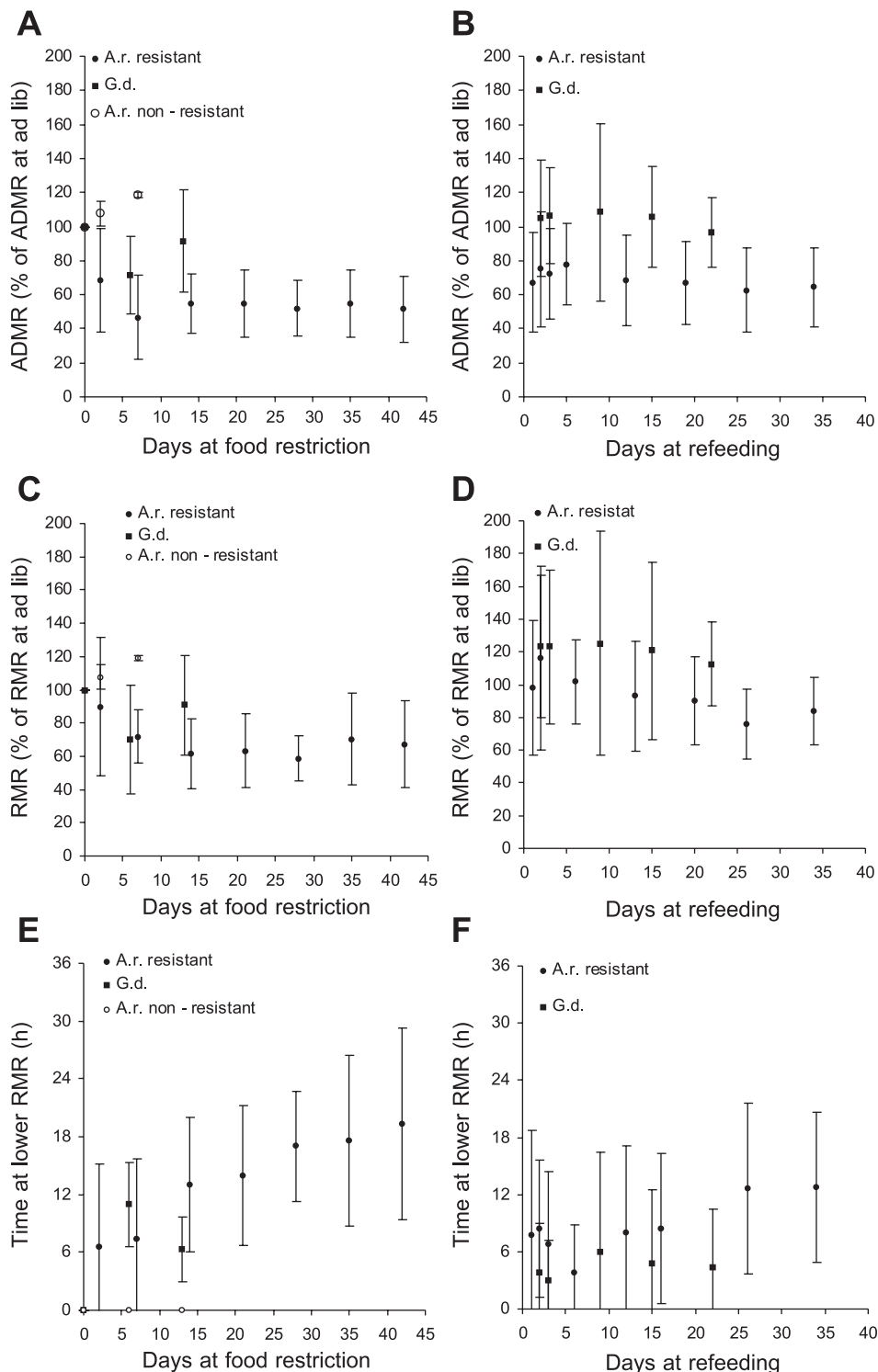


Fig. 2. Changes in average daily metabolic rate (%ad libitum ADMR, mean $\pm$ SD; A and B), resting metabolic rate (%ad libitum RMR, the lowest 20 min of the day, mean $\pm$ SD; C and D), and time (hour) spent at metabolic rate lower than ad libitum RMR (E and F) during food restriction (A, C, and E) and refeeding (B, D, and F) in resistant golden spiny mice ($n = 4$), nonresistant golden spiny mice ($n = 2$), and Wagner's gerbils ($n = 6$). Ad libitum ADMR: A.r. resistant, 0.91 (SD 0.3), A.r. nonresistant, 1.13 (SD 0.59), and G.d. 1.23 (SD 0.33) $\text{ml} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$; ad libitum RMR: A.r. resistant, 0.50 (SD 0.15), A.r. nonresistant, 0.45 (SD 0.22), and G.d. 0.85 (SD 0.39) $\text{ml} \cdot \text{g}^{-1} \cdot \text{h}^{-1}$.

the following measurements during food restriction). In non-resistant golden spiny mice, there was no significant effect of food restriction on ADMR ($P > 0.05$). In Wagner's gerbils, there was a transient decrease in ADMR after 1 wk at food restriction; however, this decrease was insignificant ($P > 0.05$).

The observed reduction in ADMR of resistant golden spiny mice during food restriction was achieved mainly by reducing RMR ($P < 0.001$; post hoc analysis between ad libitum and all measurements during food restriction, $P < 0.01$, Figs. 2C and 3), combined with a gradual but consistent and significant increase in the time spent at that lower RMR ($P < 0.05$, Fig. 2E). RMR of nonresistant golden spiny mice did not decrease significantly during food restriction ($P > 0.05$ and Figs. 2B and 3); therefore, there was no increase in time spent at lower RMR (time spent at lower RMR = 0, Fig. 2E). RMR of Wagner's gerbils did not change significantly during food restriction, even though an insignificant and transient decrease in RMR was observed after 1 wk of food restriction ($P > 0.05$, Figs. 2B and 3).

The daily increase in oxygen consumption of resistant golden spiny mice during food restriction took place mainly at and around feeding time, whereas that of the gerbils and nonresistant golden spiny mice lasted for several hours (Fig. 3).

During refeeding, ADMR of resistant golden spiny mice was significantly lower than during ad libitum ($\sim 80\%$ of their ad libitum ADMR, $P < 0.05$; post hoc analysis $P < 0.05$ for all days of refeeding compared with ad libitum), whereas in gerbils it returned to ad libitum values after 3 days of refeeding

($P > 0.05$ between ad libitum and refeeding, Fig. 2B). During refeeding, RMR was not significantly different from ad libitum RMR in both species ($P > 0.05$), and so was time spent at lower RMR ($P > 0.05$, Fig. 2F).

Core body temperature and heart rate. There was a significant effect of days at food restriction on average body temperature in both species. The effect of food restriction on average daily body temperature of both types of golden spiny mice did not differ significantly, and there was no interaction during the period when both types were under food restriction (first 12 days of food restriction, days at food restriction $P < 0.001$; type $P > 0.05$; interaction, $P > 0.05$, Fig. 4, B and D). The decrease in body temperature resulted mainly from a decrease in minimal body temperature in both types; there was a significant effect of days at food restriction on minimal body temperature values, without interaction between types of golden spiny mice (days at food restriction $P < 0.001$, type $P > 0.05$; interaction, $P > 0.05$, Fig. 4, B and D). There was no significant effect of days at food restriction on maximal body temperature and no interaction between spiny mice types during these first 12 days (days at food restriction $P > 0.05$, type $P > 0.05$, interaction $P > 0.05$, Fig. 4, B and D). The temporal pattern of body temperature during food restriction differed between golden spiny mice types; in nonresistant golden spiny mice, body temperature increased around both light onset and offset similarly to the ad libitum period, whereas that of the resistant golden spiny mice increased around dark onset only, i.e., feeding time (Fig. 5, A and C).

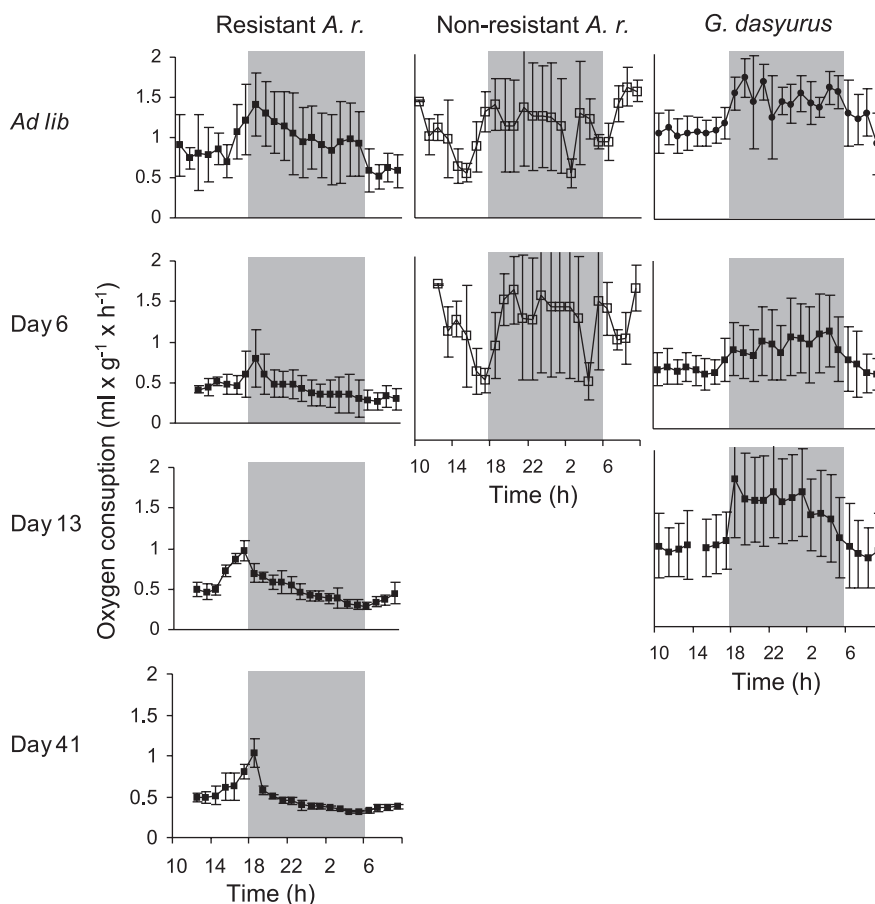


Fig. 3. Daily rhythms of oxygen consumption (mean $\pm$ SD) of resistant ($n = 4$) and nonresistant ($n = 2$) golden spiny mice and of Wagner's gerbils ($n = 6$) during ad libitum and food restriction. Dark background represents dark hours. Feeding time, 1815.

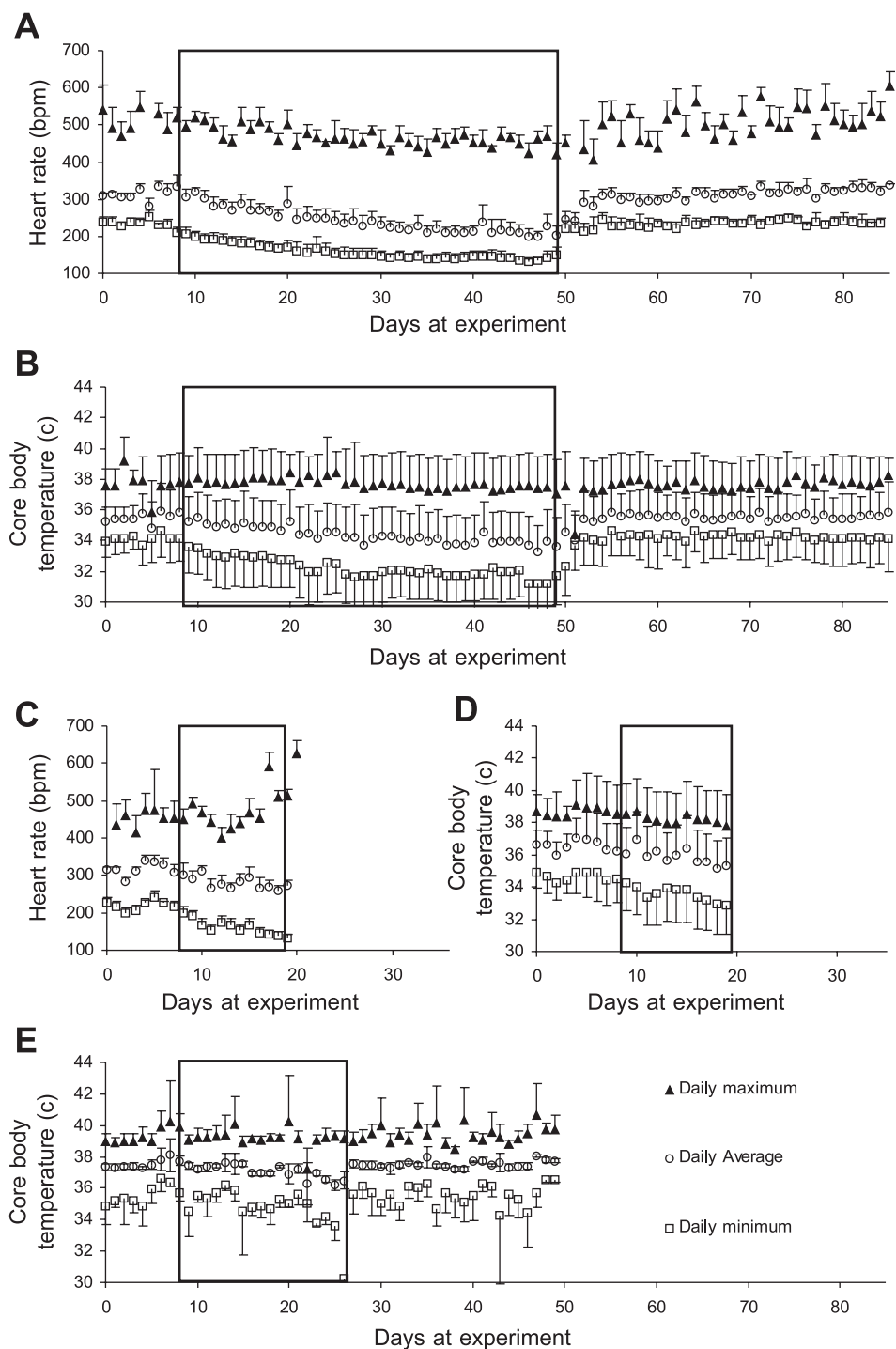


Fig. 4. Heart rate (A and C) and body temperature (B, D, and E; minimum, maximum, and daily mean $\pm$ SD) of resistant (A and B, $n = 4$) and nonresistant (C and D, $n = 2$) golden spiny mice and body temperature (minimum, maximum, and daily mean $\pm$ SD) of Wagner's gerbils (E, $n = 6$) during ad libitum, 50% food restriction, and refeeding.

Food restriction had a significant effect on heart rate of both types of spiny mice (days at food restriction $P < 0.001$, type $P > 0.05$; interaction, $P > 0.05$, Fig. 4, A and C). The decrease in heart rate resulted mainly from a decrease in minimal heart rate values in both golden spiny mice types; there was a significant difference between resistant and nonresistant mice (days at food restriction $P < 0.001$, type $P < 0.05$, interaction $P > 0.05$, Fig. 4, A and C), and no effect of food restriction or type of spiny mice on maximal heart rate, but the interaction was significant (days at food restriction $P > 0.05$, type $P > 0.05$, interaction $P < 0.05$, Fig. 4, A and C).

Comparing the golden spiny mice with the gerbils, we found that during ad libitum, average body temperature of golden spiny mice was significantly lower ($P < 0.01$). During the first 18 days of food restriction (while the gerbils were still on 50% food restriction), average body temperature of resistant golden spiny mice and Wagner's gerbils decreased significantly (days at food restriction $P < 0.001$, species $P < 0.05$, interaction $P < 0.01$, Fig. 4, B and E), albeit in a different pattern; golden spiny mice average body temperature decreased significantly after 24 h of food restriction and was significantly lower than ad libitum during all the food restriction (post hoc analysis

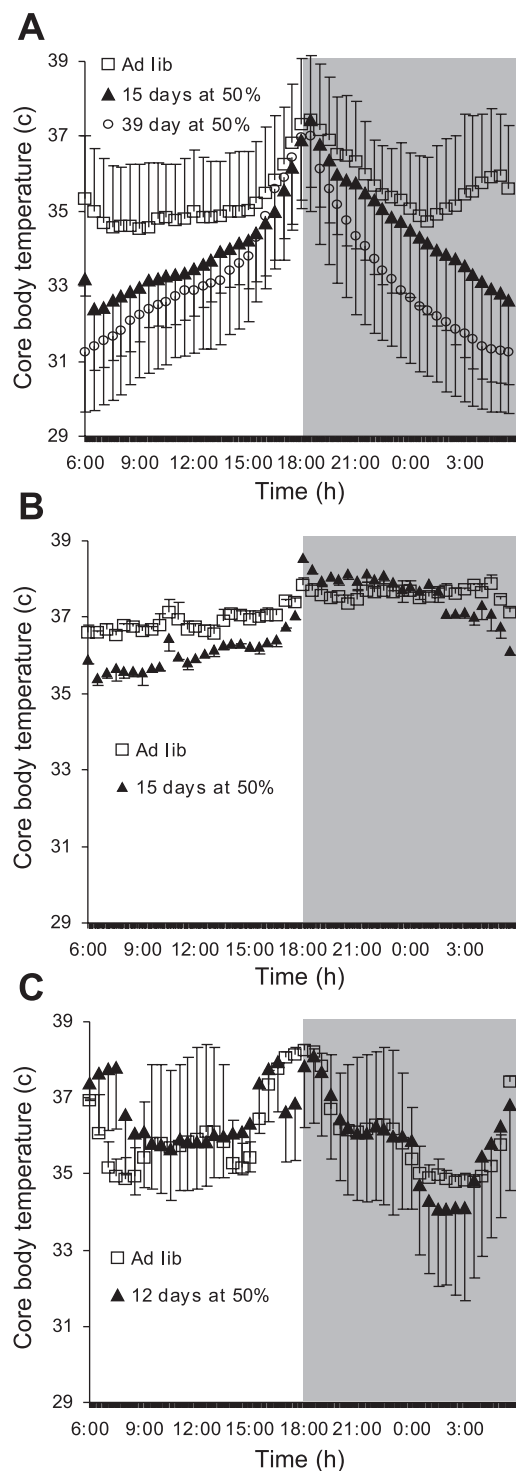


Fig. 5. Daily rhythms of body temperatures (mean $\pm$ SD) of resistant ($n = 4$, A) and non-resistant ($n = 2$, C) golden spiny mice and of Wagner's gerbils ($n = 6$, B) during ad libitum and food restriction. Dark background represents dark hours. Feeding time, 1815.

between last day at ad libitum, and all days of food restriction, $P < 0.05$). In Wagner's gerbil, average body temperature decreased significantly only after 8 days of food restriction (post hoc analysis between last day at ad libitum and all days of food restriction; $P > 0.05$ until day 8 of food restriction, $P < 0.05$ from day 8 and on, Fig. 4, B and E). As in the spiny

mice, the reduction in average body temperature of the gerbils resulted mainly from a reduction in the minimal body temperature values ($P < 0.001$, Fig. 4, B and E) without any effect on maximal body temperature values ($P > 0.05$, Fig. 4, B and E). The increase in body temperatures of resistant golden spiny mice took place mainly at and around feeding time, whereas that of the gerbils and the non-resistant golden spiny mice lasted for several hours (Fig. 5).

Leptin levels. Plasma leptin levels of resistant golden spiny mice decreased during food restriction ($P < 0.001$). The decrease was already significant after 24 h of food restriction (post hoc analysis between ad libitum and 1st day at food restriction, $P < 0.001$, Fig. 6A). In the gerbils, the decrease in plasma leptin levels was not significant after 24 h of food restriction ($P < 0.05$; post hoc analysis between ad libitum and 1st day at food restriction, $P > 0.05$, Fig. 6C). During food restriction, plasma leptin levels of golden spiny mice were not significantly correlated with body mass ($r^2 = 0.9673$, $P > 0.05$, Fig. 6A), whereas plasma leptin levels of the gerbils were highly correlated with body mass ($r^2 = 0.9398$, $P < 0.05$, Fig. 6C).

During refeeding, plasma leptin levels of golden spiny mice did not change significantly ($P > 0.05$, Fig. 6B) although they were significantly higher than during food restriction ($P < 0.01$; post hoc analysis between last measurement during food restriction and all days of refeeding measurement, $P < 0.05$, Fig. 6B), and some of the measurements were significantly lower than ad libitum leptin plasma levels (post hoc analysis between ad libitum and refeeding measurements, days 12, 26, and 40 at refeeding, $P < 0.05$, Fig. 6B). Plasma leptin levels of Wagner's gerbils changed significantly during refeeding ($P < 0.05$, Fig. 6D) and were not significantly different than at ad libitum from day 11 of refeeding (post hoc analysis between ad libitum and refeeding measurements, $P > 0.05$, Fig. 6D). During refeeding, the correlation between plasma leptin levels and body mass of golden spiny mice was very weak ($r^2 = 0.0144$), whereas a much higher correlation was found in Wagner's gerbils ($r^2 = 0.6606$, Fig. 6, B and D).

DISCUSSION

As we had predicted, most of the food-restricted golden spiny mice were able to defend their body mass (after an initial decrease), as opposed to Wagner's gerbils (Fig. 1A). In fact, golden spiny mice maintained a balanced energy budget on 50% food restriction and appeared to be able to withstand food restriction for a long period of time. These findings are in agreement with our hypothesis that Wagner's gerbils, which create seed caches in their burrows to be consumed during food shortage periods, would be less adapted physiologically to coping with food shortage, whereas golden spiny mice do not store food and are thus expected to be more adapted physiologically to long periods of food shortage in their natural habitat.

Two out of the 12 golden spiny mice tested lost weight at a higher rate than the others (Fig. 1A). A similar response to food restriction of golden spiny mice was found by Rubal (31). In his experiment, two out of six golden spiny mice did not survive food restriction and died after losing $\sim 30\%$ of their initial body mass (31). Therefore, when these two golden spiny mice lost 25% of their initial body mass, we terminated their

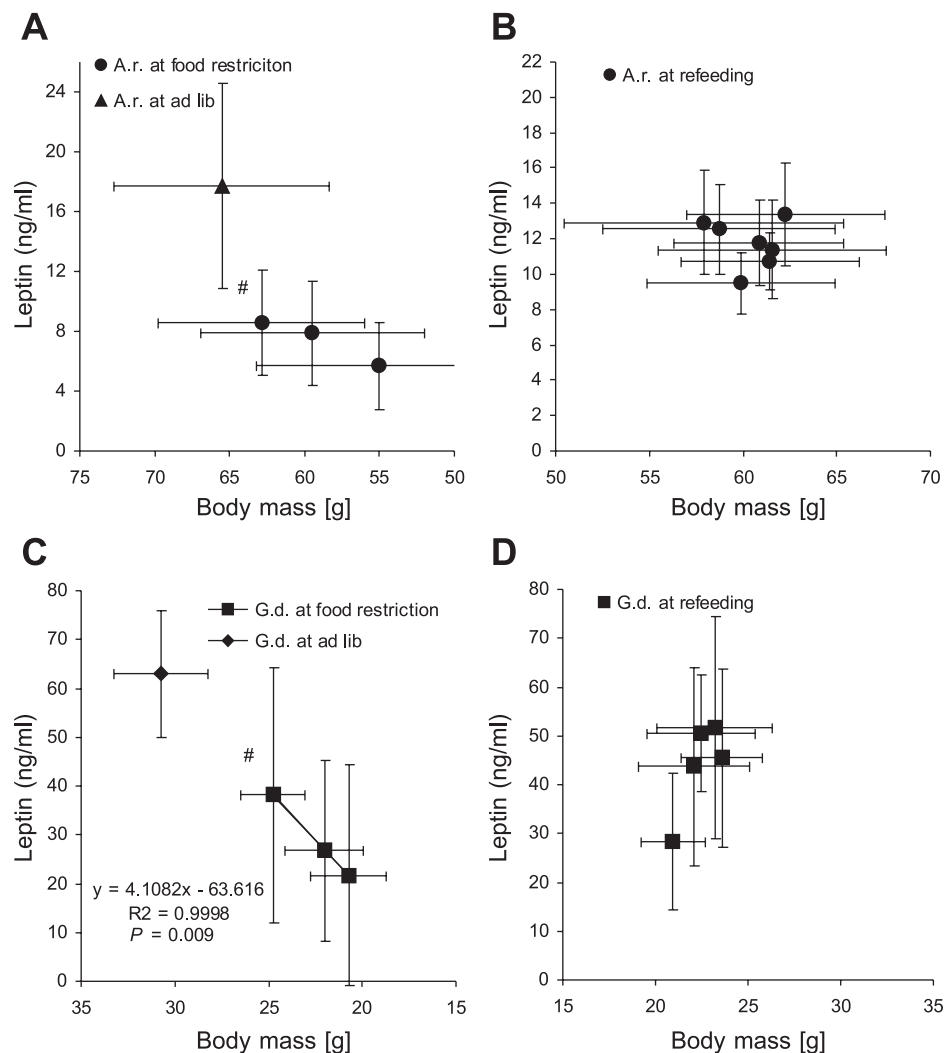


Fig. 6. Plasma leptin levels (mean $\pm$ SD) during ad libitum, food restriction (A and C) and refeeding (B and D) in resistant golden spiny mice (A and B, $n = 4$) and Wagner's gerbils (C and D, $n = 6$). #First day at food restriction.

food restriction period. These nonresistant individuals did not reduce their RMR and ADMR rate, unlike the other golden spiny mice individuals, even though they used daily torpor, as did the resistant golden spiny mice. It is possible that these two individuals had increased their general activity, as previously documented in several other species [e.g., lemurs (13) and mice (48)], or displayed starvation-induced hyperactivity, as observed in numerous species in the presence of a running wheel (reviewed in Refs. 22 and 40). We recently found in another experiment we conducted that nonresistant golden spiny mice (5 out of 12 tested) increased their general activity in response to food restriction, whereas the resistant golden spiny mice (7 out of the 12 tested) decreased it (Gutman and Kronfeld-Schor, unpublished observation).

Resistant golden spiny mice maintained body mass constant (after the initial drop) during food restriction by balancing energy expenditure against their limited energy intake (Fig. 2A). A significant decrease in ADMR and RMR was observed in the resistant golden spiny mice as a result of food restriction, as previously reported for both desert and nondesert mammals (5, 23, 25, 42). In Wagner's gerbils, a transient, insignificant decrease in energy expenditure was observed, and ADMR

returned to its ad libitum values after 13 days of food restriction, resulting in their inability to balance their energy budget under long food restriction. Such a transient effect of food restriction on ADMR was previously described in other species, e.g., white mice (25) and rats (23).

The observed decrease in ADMR of resistant golden spiny mice resulted from a further reduction of their already low RMR to even lower levels, an increase in the time spent at this new RMR, as similarly observed by Merkt and Taylor (25), and a reduction in the maximal values. In the Merkt and Taylor (25) experiment, a "metabolic switch" to desert survival was described in golden spiny mice; as a result of 50% food restriction, the animals decreased their mass-specific metabolic rates by $\sim 10\%$ during the first 2 wk of food restriction and then suddenly switched down RMRs by one-half in a single day (not on the same day in all individuals, average of 16 ± 3.6 days). No such phenomenon was observed in the present study.

It is important to note that the golden spiny mice used in the study of Merkt and Taylor (25) were obese, with average body mass being 85.7 ± 2.1 g ($n = 4$, 3 males and 1 female), whereas the average adult body mass of this species in the wild is 44.1 ± 0.61 g (38). In our experiments, average body mass

was 66.2 (SD 12.1) g ($n = 12$), which is also overweight, but less severely so. In fact, the observed metabolic switch in the Merkt and Taylor (25) study occurred after the animals had lost $\sim 20\%$ of their body mass, i.e., had reached a weight similar to that of our spiny mice at the beginning of the experiment, so it is possible that our mice were already below that switch point at that point. In Siberian hamsters, food restriction induces torpor only after body mass has decreased below a critical level (32, 33), and it may be that this critical level is ~ 70 g in golden spiny mice.

A major route for energy expenditure is the energy used for thermoregulation. Keeping body temperature at a constant level entails high energetic cost. In times of low energy availability, allowing body temperature to drop can reduce this cost. Body temperature of both types of golden spiny mice was significantly lower than that of Wagner's gerbils under ad libitum food availability because of a lower resting body temperature and fewer hours spent at maximal body temperature. When fed ad libitum, body temperature amplitude was $\sim 4^\circ\text{C}$ in the resistant and nonresistant golden spiny mice and $< 3^\circ\text{C}$ in Wagner's gerbils. During food restriction, both species reduced their body temperature. However, body temperature of the resistant golden spiny mice dropped within 1 day of food restriction (as did body temperature of the nonresistant animals, although the decrease was insignificant), whereas that of Wagner's gerbils dropped significantly only after 8 days of food restriction. In both species, the reduction was achieved by lowering the minimal but not the maximal body temperature; and resistant golden spiny mice spent fewer hours at maximal body temperature, increasing body temperature just before and at feeding time. In Wagner's gerbils and nonresistant golden spiny mice, body temperatures remained high throughout most of the night.

The largest drop in body temperature of resistant and non-resistant golden spiny mice occurred during daytime, decreasing to $1\text{--}2^\circ\text{C}$ above ambient temperature (30°C). These drops can be described as hypothermia or torpor. Torpor is not a precisely defined term but is generally used for a variety of states in which metabolic rate, during a part of the circadian cycle, falls below its normal resting level, permitting body temperature to approach the ambient temperature (9). Torpor has been described in a variety of small mammals, mostly from cold habitats (e.g., Ref. 35, but see 2). Usually, torpor is induced by and studied under low ambient temperature (e.g., 9, 10, 35). However, other triggers, including reduced food availability, can induce torpor (e.g., 3, 13, 34, 36, 44).

The ability of the resistant golden spiny mice to reduce their energy expenditure, at least partially by lowering their body temperature, compared with Wagner's gerbils, and the differences in the responses of the resistant and nonresistant golden spiny mice, may have resulted from natural selection in their natural habitat, and in relation to their diet. Because the two species used in this study originated from laboratory colonies that are raised under ad libitum food supply, this ability could not have resulted from prior acclimatization to different conditions, or from gestational programming (30), but, rather, constitutes a genetic character of the species.

The fact that 2 out of the 12 golden spiny mice individuals tested showed a different response to food restriction may indicate that this species uses one of two different strategies to

cope with food shortage, as previously described in *Phodopus sungorus* (35): increased foraging or migration to a better habitat (a strategy that we are currently studying); or lowering energy demands by either entering daily torpor (as we have shown in this experiment) and/or lowering activity level. Such individual variation may have an ecological advantage, since it allows a flexible response of the population toward fluctuations in food availability.

The hormone leptin may be involved in the way food consumption influences thermoregulatory mechanisms and metabolic rate. A role of leptin in the response to fasting was first suggested by Ahima et al. (1). Leptin is a hormone secreted mainly by white fat cells, and its concentration in the plasma generally correlates with body fat mass. During food shortage periods, animals use their fat stores as an energy source. This causes a reduction in fat mass, and eventually a reduction in plasma leptin concentration. The decreased leptin levels and subsequent decreased signaling in the hypothalamus may be a crucial mediator of many of the neuroendocrine and hypothalamic responses to fasting (1, 9).

It has been suggested that, when food is scarce, the primary role of leptin is in the control of thermoregulatory energy expenditure and that it plays a crucial role in determining the frequency and/or magnitude of the extent of the decreases in metabolic rates and thus the extent of utilization of the endogenous energy stores (9, 12, 11). Accordingly, in hibernating little brown bats, leptin concentrations decrease despite an increase in fat stores, possibly to remove their inhibitory effect on hibernation (20).

In the current study, we found that resistant golden spiny mice leptin levels dropped significantly and to a higher extent than expected after 24 h of food restriction (Fig. 6). This may have been the signal that stimulated the rapid drop in metabolic rate observed after 1 wk of food restriction and allowed the onset of torpor. Leptin levels continued to decrease with a high leptin-fat correlation during food restriction, presumably allowing tight control of energy expenditure during this time. During refeeding, the correlation between plasma leptin level and fat content was low. Golden spiny mice gained fat, but their leptin levels remained relatively constant. Unfortunately, we did not collect blood samples from the two nonresistant golden spiny mice. In Wagner's gerbils, the correlation between plasma leptin concentration and body mass remained relatively high throughout the experiment.

In food-restricted Sprague-Dawley rats, plasma leptin level returned to the basal level after only 2 days of refeeding and was equivalent to that of nonrestricted control rats. It continued to increase, even though the rat's body mass remained significantly lower than that of nonrestricted controls (45). Similar results were documented in children during the period of catch-up growth (4). Furthermore, leptin treatment during refeeding reduced food intake in rats (45), and in the marsupial *Minthopsis crassicaudata* (51). It is possible that the low plasma leptin levels and the low correlation between body fat and plasma leptin levels found in golden spiny mice during refeeding in the current study allowed them to continue eating and gain fat.

In summary, most golden spiny mice showed a remarkable ability to balance their energy budget throughout a period of 50% food restriction and to maintain their body mass constant

after an initial drop during chronic caloric restriction. This ability was achieved at least partly by lowering their body temperature, thereby reducing thermoregulatory energy expenditure. The mechanism by which food availability influences thermoregulatory mechanisms and metabolic rate may involve the hormone leptin. Furthermore, it is possible that golden spiny mice use two different strategies for coping with food shortage periods. This hypothesis should be further studied in this species.

ACKNOWLEDGMENTS

We thank M. Meir for technical help, E. Shargal for assistance with the golden spiny mice experiment, O. Shayovich and J. Weisemberg for assistance with the Wagner's gerbils experiment, I. R. Khokhlova and B. R. Krasnov from the Ramon Science Center for kindly providing the Wagner's gerbils, I. Gelernter from the statistical laboratory at Tel Aviv University for statistical advice, and N. Paz for scientific editing. We also thank E. P. Widmaier and the two anonymous reviewers for enlightening comments on previous versions of this manuscript.

REFERENCES

- Ahima RS, Prabakaran D, Mantzoros C, Qu D, Lolli B, Maratos-Flier E, and Flier JS. Role of leptin in the neuroendocrine response to fasting. *Nature* 382: 250–252, 1996.
- Bozinovic F and Marquet PA. Energetics and torpor in the atacam desert-dwelling rodent *Phyllotis darwini rupestris*. *JAMA* 72: 734–738, 1991.
- Brown JH and Bartholomew GA. Periodicity and energetics of torpor in the kangaroo mouse, *Microdipodops pallidus*. *Ecology* 50: 705–709, 1969.
- Buyukgebiz B, Ozturk Y, Yilmaz S, and Arslan N. Serum leptin concentration in children with mild protein-energy malnutrition and catch-up growth. *Pediatrics* 46: 534–538, 2004.
- Choshniak I, Ben-Kohav N, Taylor CR, Robertshaw D, Barnes RJ, Dobso A, Belkin V, and Shkolnik A. Metabolic adaptation for desert survival in the Bedouin goat. *Am J Physiol Regul Integr Comp Physiol* 268: R1101–R1110, 1995.
- Degen AA. *Ecophysiology of Small Desert Mammals*. Berlin, Germany: Springer, 1997.
- Degen AA and Kam M. Water intake in two coexisting desert rodents, *Acomys cahirinus* and *Gerbillus dasyurus*. *J Mammal* 73: 201–206, 1992.
- Depocas F and Hart JS. Use of the Pauling oxygen analyzer for measurements of oxygen consumption of mammals in open-circuit system and in short-lag closed circuit apparatus. *J Appl Physiol* 20: 261–263, 1957.
- Doring H, Schwarzer K, Nuesslein-Hildesheim B, and Schmidt I. Leptin selectively increases energy expenditure of food-restricted lean mice. *Int J Obes* 22: 83–88, 1998.
- Elliot JA, Barteness TJ, and Goldman BD. Role of short photoperiod and cold exposure in regulating daily torpor in Djungarian hamsters. *J Comp Physiol [A]* 161: 245–253, 1987.
- Freeman DA, Lewis DA, Kauffman AS, Blum RM, and Dark J. Reduced leptin concentrations are permissive for display of torpor in Siberian hamsters. *Am J Physiol Regul Integr Comp Physiol* 287: R97–R103, 2004.
- Geiser F, Kortner G, and Schmidt I. Leptin increase energy expenditure of a marsupial by inhibition of daily torpor. *Am J Physiol Regul Integr Comp Physiol* 275: R1627–R1632, 1998.
- Genin F and Perret M. Daily hypothermia in captive grey mouse lemurs (*Microcebus murinus*): effects of photoperiod and food restriction. *Comp Biochem Physiol B* 136: 71–81, 2003.
- Grodzinski W and Wunder BA. Ecological energetics of small mammals. In: *Small Mammals their Productivity and Population Dynamics*, edited by Golley FB, Petruszewicz K, and Ryszkowski L. Cambridge UK: Cambridge Univ Press, 1975, p. 541.
- Haim A. Thermoregulation and metabolism of Wagner's gerbil (*Gerbillus dasyurus*): a rock dwelling rodent adapted to arid and mesic environments. *J Therm Biol* 12: 45–48, 1987.
- Hatough-Bouran A. The burrowing habits of desertic rodent *Jaculus jaculus* and *Gerbillus dasyurus* in the Shaumari Reserve in Jordan. *Mammalia* 54: 341–359, 1990.
- Kinlaw A. A review of borrowing by semi-fossorial vertebrates in arid environments. *J Arid Environ* 41: 127–145, 1999.
- Kleiber M. *The Fire of Life*. New York, NY: Wiley, 1961.
- Kronfeld N and Dayan T. The dietary basis for temporal partitioning: food habits of coexisting *Acomys* species. *Oecologia (Berl)* 121: 123–128, 1999.
- Kronfeld-Schor N, Silvia AB, Kunz TH, Richardson C, and Widmaier EP. Dissociation of leptin secretion and adiposity during pre-hibernation fattening in little brown bats. *Am J Physiol Regul Integr Comp Physiol* 279: R1277–R1281, 2000.
- Louw G and Seely M. *Ecology of Desert Organisms*. London, UK: Longman, 1982.
- Mather JG. Wheel-running activity: a new interpretation. *Mamm Rev* 11: 41–51, 1981.
- McCarter RJ and McGee JR. Transient reduction of metabolic-rate by food restriction. *Am J Physiol Endocrinol Metab* 257: E175–E179, 1989.
- Mendelssohn H and Yom-Tov Y. *Fauna Palestina: Mammalia of Israel*. Israel: Israel Acad Sci Human, 1999.
- Merk J and Taylor CR. A metabolic switch for desert survival. *Proc Natl Acad Sci USA* 91: 12313–12316, 1994.
- Nagy TR and Anne-Laure C. Precision and accuracy of dual-energy X-ray absorptiometry for determining in vivo body composition of mice. *Obesity Res* 8: 392–398, 2000.
- Nowak RM. *Walker's Mammals of the World*. Baltimore, MD: Johns Hopkins Univ Press, 1999.
- Price VM and Joyner JW. What resource are available to desert granivores: seed rain or soil seed bank? *Ecology* 78: 764–773, 1997.
- Reichman OJ, Wicklow DT, and Rebar C. Ecological and mycological characteristics of caches in the mounds of *Dipodomys spectabilis*. *J Mammal* 66: 643–651, 1985.
- Ross MG and Desai M. Gestational programming: population survival effects of drought and famine during pregnancy. *Am J Physiol Regul Integr Comp Physiol* 288: R25–R33, 2005.
- Rubal A. *Energy Metabolism: Adaptation for Life in the Desert*. Tel Aviv, Israel: Tel Aviv Univ, 1989.
- Ruby NF. Paraventricular nucleus ablation disrupts daily torpor in Siberian hamsters. *Brain Res Bull* 37: 193–198, 1995.
- Ruby NF, Nelson RJ, Licht P, and Zucker I. Prolactin and testosterone inhibit torpor in Siberian hamsters. *Am J Physiol Regul Integr Comp Physiol* 264: R123–R128, 1993.
- Ruby NF and Zucker I. Daily torpor in the absence of the suprachiasmatic nucleus in siberian hamsters. *Am J Physiol Regul Integr Comp Physiol* 263: R353–R362, 1992.
- Ruf T, Klingenspor M, Preis H, and Heldmaier G. Daily torpor in the Djungarian hamsters (*Phodopus sungorus*): interactions with food-intake, activity, and social-behavior. *J Comp Physiol [B]* 160: 609–615, 1991.
- Ruf T, Stieglitz A, Steinlechner S, Blank JL, and Heldmaier G. Cold exposure and food restriction facilitate physiological responses to short photoperiod in Djungarian hamsters (*Phodopus sungorus*). *J Exp Zool* 267: 104–112, 1993.
- Schmidt-Nielsen K. *Animal Physiology Adaptation and Environment*. Cambridge, UK: Cambridge Univ Press, 1997.
- Shargal E, Kronfeld-Schor N, and Dayan T. Population biology and spatial relationships of coexisting spiny mice of the genus *Acomys*. *J Mammal* 81: 1046–1052, 2000.
- Shenbrot GI, Krasnov BR, and Khokhlova IS. Biology of Wagner's gerbil *Gerbillus dasyurus* (Wagner, 1842) (Rodentia: Gerbillidae) in the Negev Highlands, Israel. *Mammalia* 61: 467–486, 1997.
- Sherwin CM. Voluntary wheel running: a review and novel interpretation. *Anim Behav* 56: 11–27, 1998.
- Shkolnik A. *Studies in the Comparative Biology of Israel's Two Species of Spiny Mice (genus Acomys)* (PhD thesis). Israel: Hebrew Univ, 1996.
- Shkolnik A. Physiological adaptation to the environment: the Israeli experience. In: *The Zoogeography of Israel*, edited by Yom-Tov Y and Tchernov E. Dordrecht, Holland: Dr. W. Junk, 1988.

43. **Shkolnik A and Borut A.** Temperature and water relations in two species of spiny mice (*Acomys*). *J Mammal* 50: 245–255, 1969.
44. **Tannenbaum MG and Pivorun EB.** Differential effect of food restriction on the induction of daily torpor in *Peromyscus maniculatus* and *Peromyscus leucopus*. *J Therm Biol* 12: 159–162, 1987.
45. **Velkoska E, Morris MJ, Burns P, and Weisinger RS.** Leptin reduces food intake but does not alter weight regain following food deprivation in the rat. *Int J Obes* 27: 48–54, 2003.
46. **Weinstein M.** *Spiny Arthropods, Spiny Parasites, and Spiny Mice: Availability and Ecology in En Gedi*. Tel Aviv, Israel: Tel Aviv Univ, 2004.
47. **Weinstein M, Dayan T, and Kronfeld-Schor N.** Characterizing an arthropod community in relation to spiny mouse diets (Abstract). *Isr J Zool* 50: 124, 2004.
48. **Williams TD, Chambers JB, Henderson RP, Rashotte ME, and Overton JM.** Cardiovascular responses to caloric restriction and thermoneutrality in C57BL/6J mice. *Am J Physiol Regul Integr Comp Physiol* 282: R1459–R1467, 2002.
49. **Willmer PJJ, Johnston IA, and Stone G.** *Environmental Physiology of Animals*. Madlen, MA: Blackwell, 2000, p. 644.
50. **Withers PC.** Measurements of $\dot{V}O_2$, $\dot{V}CO_2$, and evaporative water loss with a flow-through mask. *J Appl Physiol Respir Environ Exercise Physiol* 42: 120–123, 1977.
51. **Wittert GA, Turnbull H, Hope P, Morley JE, and Horowitz M.** Leptin prevents obesity induced by a high-fat diet after diet-induced weight loss in the marsupial *S. crassicaudata*. *Am J Physiol Regul Integr Comp Physiol* 286: R734–R739, 2004.

